

WEEK 2



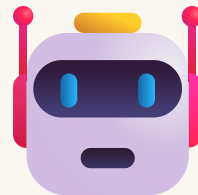
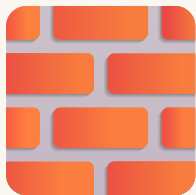
Turning at Walls

Wall ahead? Turn. Path clear? Go.

벽이 있으면 돌고, 길이 열리면 전진 (영어코딩)

BIG PICTURE

One rule, repeated forever



If a wall is ahead, turn. Otherwise, step forward.

한 가지 규칙을 끝없이 반복해요

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

`if` / `else` — pick a branch

The agent asks one question, then runs **one** of two paths.

```
if wall ahead:  
    turn right  
else:  
    move forward
```

Only one branch runs each time — never both.

조건에 따라 둘 중 하나만 실행돼요

WATCH IT!

Wall → turn · clear → move

wall ahead

path clear

`agent.detect(BLOCK, FORWARD)?`

✓ `true` → `agent.turn(RIGHT_TURN)`

✗ `false` → `agent.move(FORWARD, 1)`

▸ Agent does

CODE

The turning rule

```
if agent.detect(AgentDetection.BLOCK, FORWARD):  
    agent.turn(RIGHT_TURN)  
else:  
    agent.move(FORWARD, 1)
```

Wall in front → spin right. Clear → take one step.

벽이면 오른쪽으로, 아니면 한 칸 전진

CODE

Repeat it to solve the maze

```
while not agent.detect(AgentDetection.BLOCK, DOWN):  
    if agent.detect(AgentDetection.BLOCK, FORWARD):  
        agent.turn(RIGHT_TURN)  
    else:  
        agent.move(FORWARD, 1)
```

A `while` loop wraps the rule so it runs again and again.

CONCEPT

Choosing a turn with `elif`

```
direction = "right"

if direction == "left":
    agent.turn(LEFT_TURN)
elif direction == "right":
    agent.turn(RIGHT_TURN)
else:
    agent.move(FORWARD, 1)
```

`elif` = "else, if instead...". Checked only if the first was false.

PREDICT 🤖

What will the agent do?

Predict first — then click reveal

```
if agent.detect(BLOCK, FORWARD):  
    agent.turn(RIGHT_TURN)  
else:  
    agent.move(FORWARD, 1)
```

Reveal output 🔍 reveal

VOCAB

New words today

if / else run one branch when true, the other when false

branch one of the two paths the code can take

elif check another condition if the first was false

turn change facing without changing spot

조건문

분기 (갈래)

아니면 만약

돌다



What's wrong with this code?

Goal: turn when blocked, move when clear.

buggy

```
if not agent.detect(BLOCK, FORWARD):  
    agent.turn(RIGHT_TURN)  
else:  
    agent.move(FORWARD, 1)
```

Think first — then click next.



DEBUG

The bug

Bug: the test is inverted. It turns when the *path is clear* and walks *into* walls. Remove `not` so it turns when a wall is detected.

fixed

```
if agent.detect(BLOCK, FORWARD):  
    agent.turn(RIGHT_TURN)  
else:  
    agent.move(FORWARD, 1)
```

YOUR TURN!

Turn through the maze

✓ Open the Week 2 world + Code Builder

✓ Write the `if` / `else` turning rule

✓ Wrap it in a `while` loop

✓ Guide the agent to the goal

✓ Send a photo or video on KakaoTalk



QUIZ

QUESTION 1 OF 3

In an `if / else`, how many branches run each time?

Both branches always run.

It depends on the loop count.

Exactly one — true runs the if, false runs the else.

Neither runs.



QUIZ

QUESTION 2 OF 3

When is an `elif` branch checked?

Always, before the if.

Only if the earlier if/elif conditions were false.

Only inside a while loop.

Never — it's ignored.



QUIZ

QUESTION 3 OF 3

What's the difference between **turn** and **move** ?

They do the same thing.

move only works downward.

turn places a block; move removes one.

turn changes facing in place; move walks to a new block.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↶ retake



Turning solved!

Next: check two things at once with AND.

다음 주: AND로 두 가지를 한 번에 확인